SMART OBSTACLE STOP ROBOT

Assignment 4 — Robotics Operating Systems (ROS 2 Jazzy, Python)

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Package Name: obstacle_robot

Ros2 Distribution: Jazzy

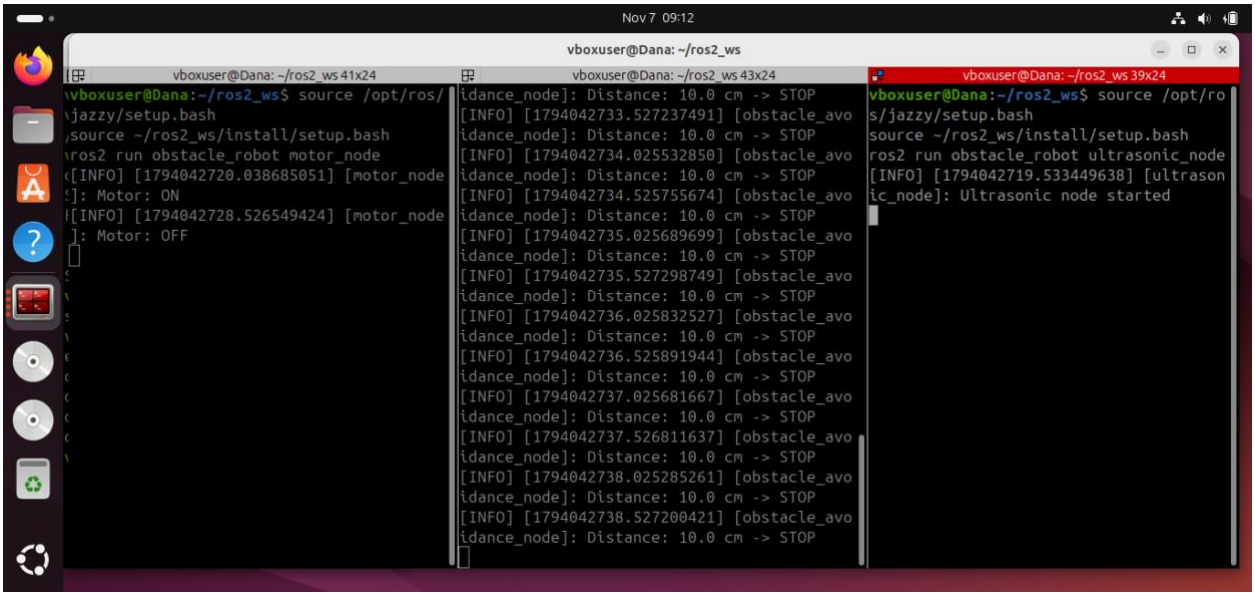
Programming language: Python

This assignment implements a three-node ROS2 system that monitors an obstacle distance , selects a motor command , and reports the motor state .

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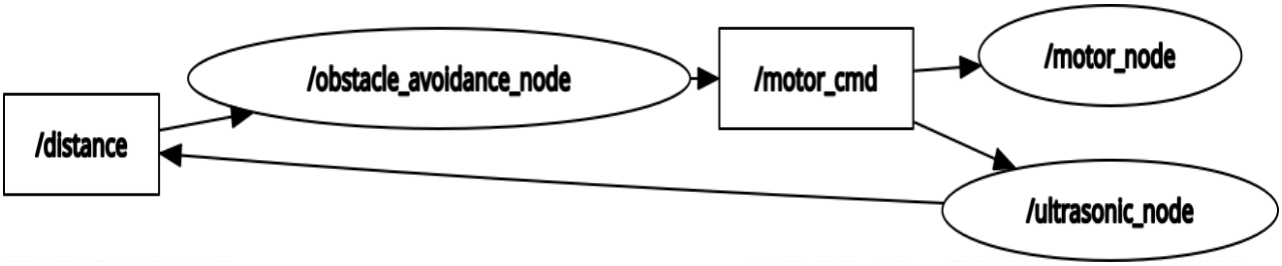
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The three terminals show the motor node, obstacle-avoidance node, and ultrasonic node running at the same time. The motor changes from ON to OFF, and the decision node repeatedly reports STOP When the measured distance is 10.0 cm .



Screenshot 1. The three ROS 2 nodes are running in separate terminals. The motor output shows the transition from ON to OFF , While the obstacle-avoidance node shows the STOP command at a distance of 10.0 cm .

The rqt_graph visualisation confirms the communication connections between the three nodes. The ultrasonic node publishes / distance to the obstacle-avoidance node, which publishes/motor_cmd to both the motor node and the ultrasonic node.



Screenshot 2. The rqt_graph shows all three nodes and the two topics used for communication: /distance and /motor_cmd.

Conclusion: The ROS 2 Graph demonstrates the required publisher-subscriber structure, and the terminal output shows that the robot stops when the obstacle distance is below 20.0 cm.